Lecture-03

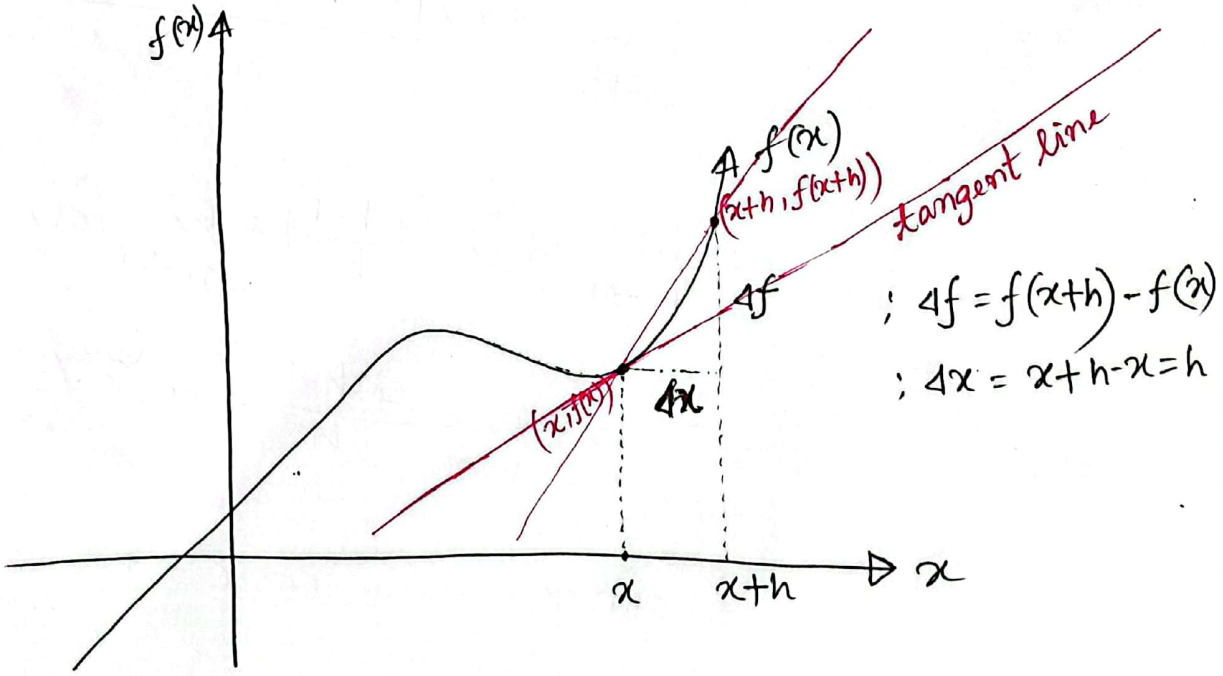
★ Differentiability:

A function $f(x)$ is said to be differentiable at $x = c$ if $\lim_{h \to 0} \frac{f(c+h) - f(c)}{h}$ exists.

The function $f'(x)$, defined by the form,

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

is called the derivative of f with respect to x. The domain of $f'(x)$ consists of all x in the domain of f for which the limit exists.



* **Example:** Use the limit definition to find the derivative of $f(x) = \sin x$ at $x = \frac{\pi}{2}$.

Solution

The derivative of the function using limit definition can be written as,

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \to 0} \frac{\sin(x+h) - \sin(x)}{h}$$

$$= \lim_{h \to 0} \frac{\sin x \cos h + \cos x \sin h - \sin x}{h}$$

$$= \lim_{h \to 0} \frac{\sin x (\cos h - 1) + \cos x \sin h}{h}$$

$$= \lim_{h \to 0} \sin x \left(\frac{\cos h - 1}{h}\right) + \lim_{h \to 0} \cos x \frac{\sin h}{h}$$

$$= \sin x \cdot \lim_{h \to 0} \frac{\cos h - 1}{h} + \cos x \cdot \lim_{h \to 0} \frac{\sin h}{h}$$

$$= \sin x \times 0 + \cos x \cdot 1$$

$$= \cos x$$

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$$\therefore \; f'(x) = \frac{df}{dx} = \frac{d}{dx}\sin x = \cos x$$

at $x = \frac{\pi}{2}$; $f'\left(\frac{\pi}{2}\right) = \cos(\pi/2) = 0$

Ans.

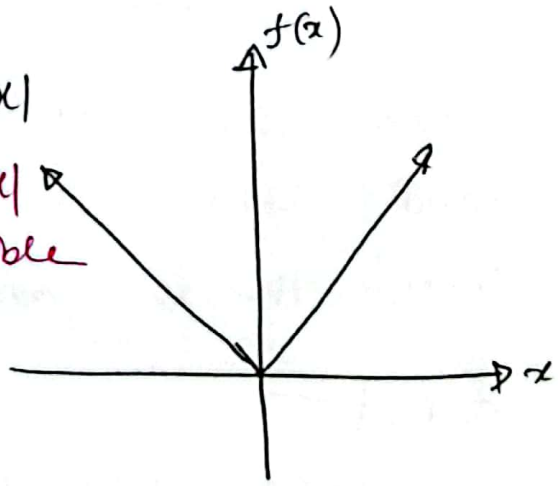
* <u>Homework</u>: Find the derivative of $f(x) = \sqrt{x}$ at $x = 9$.

* A function is not differentiable at

i) corner points → At corner points the slope of the secant lines have different limits from the left and from the right and hence the two sided limit that defines the derivative does not exist.

<u>Example</u>: $f(x) = |x|$

prove that $f(x) = |x|$ is not differentiable at $x = 0$.



$\boxed{\text{solution}}$

The derivative of $f(x)$ is defined by

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$$\Rightarrow f'(0) = \lim_{h \to 0} \frac{f(0+h) - f(0)}{h}$$

$$= \lim_{h \to 0} \frac{|h| - 0}{h}$$

$$\therefore \quad = \lim_{h \to 0} \frac{|h|}{h}$$

Lets calculate the left hand and right hand limit.

$$Lf'(0) = \lim_{h \to 0^-} \frac{|h|}{h} = \lim_{h \to 0^-} \frac{-h}{h} = -1$$

$$Rf'(0) = \lim_{h \to 0^+} \frac{|h|}{h} = \lim_{h \to 0^+} \frac{h}{h} = 1$$

since the left hand ~~deri~~ and right hand limits of $f'(0)$ are not equal, then the limit $\lim_{h \to 0} \frac{|h|}{h}$ does not exist.

Therefore, $f(x) = |x|$ is not differentiable at $x = 0$.

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ii) vertical tangency → At a point of vertical tangency the slope of the secant line approach $+\infty$ or $-\infty$ from the left and the right.

<u>Example:</u> show that $f(x) = \tan x$ is not differentiable at $x = \pi/2$.

<u>Ans:</u>

$$L.H.L = \lim_{x \to \frac{\pi}{2}^-} \tan x = \tan\left(\frac{\pi}{2}\right)$$

$$= +\infty \; ; \quad [\tan x \geq 0 \text{ at } 0 \leq x \leq \pi/2]$$

$$R.H.L = \lim_{x \to \frac{\pi}{2}^+} \tan x = -\tan \pi/2 = -\infty,$$

$$[\tan x \leq 0 \text{ ; at } \pi/2 < x \leq \pi]$$

so, $\lim_{x \to \pi/2} \tan x$ is not defined and not exists. so, $f(x) = \tan x$ is not continuous at $x = \pi/2$. hence not differentiable at $x = \pi/2$.

Example: show that

$$f(x) = \begin{cases} x^2+2 & ; x \le 1 \\ x+2 & ; x > 1 \end{cases}$$

is continuous but not differentiable at $x=1$.

$\boxed{\text{Ans:}}$

At $x=1$; $f(1) = 1^2+2 = 3$

and $\lim_{x\to 1^-} f(x) = \lim_{x\to 1^-} x^2+2 = 1+2 = 3$

$\lim_{x\to 1^+} f(x) = \lim_{x\to 1^+} (x+2) = 1+2 = 3$

since, $\lim_{x\to 1} f(x) = 3 = f(1)$

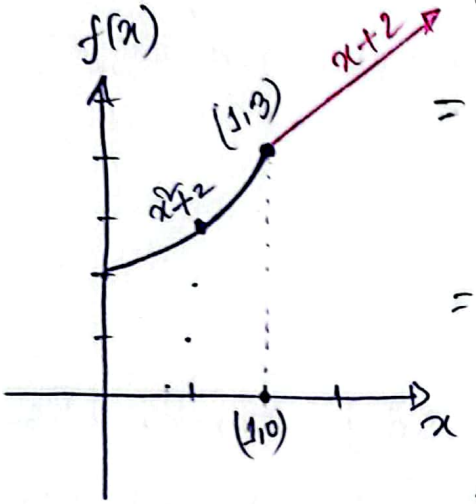
so, $f(x)$ is continuous at $x=1$

To show the differentiability of the function $f(x)$ at $x=1$, we set

$$f'(1) = \lim_{h\to 0} \frac{f(1+h) - f(1)}{h}$$

since, the behavior of the function changes at $x=1$, we have to calculate the left hand and right hand limit at $x=1$

$$Lf'(1) = \lim_{h \to 0^-} \frac{f(1+h)-f(1)}{h} \qquad ; f(1)=3$$



$$= \lim_{h \to 0^-} \frac{\{(1+h)^2+2\}-3}{h}$$

$$= \lim_{h \to 0^-} \frac{2h+h^2}{h} = \lim_{h \to 0^-} (2+h)$$

$$= 2+0 = 2$$

$$Rf'(1) = \lim_{h \to 0^+} \frac{f(1+h)-f(1)}{h}$$

$$= \lim_{h \to 0^+} \frac{\{(1+h)+2\}-3}{h}$$

$$= \lim_{h \to 0^+} \frac{h}{h} = 1$$

since $Lf'(1) \neq Rf'(1)$. so the function is not differentiable.

(Showed)

* Techniques of Differentiation:

We should learn basic differentiation of some known functions from the reference book. Then, we will try to ~~get~~ differentiate any functions using basic derivatives and differential properties.

* Theorem:

If a function f is differentiable at x_0, then f is continuous at x_0. If f is continuous at x_0, then it can not be said that the function is differentiable at x_0.

* Rules of Differentiation:

sum Rule: $\frac{d}{dx}\left(Af(x)+Bg(x)\right) = A\frac{df}{dx} + B\frac{dg}{dx}$

product Rule: $\frac{d}{dx}\left(f(x)\cdot g(x)\right) = f(x)\frac{d}{dx}g(x) + g(x)\frac{d}{dx}f(x)$

Quotient Rule: $\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{g(x)\frac{d}{dx}f(x) - f(x)\frac{d}{dx}g(x)}{(g(x))^2}$

where, $g(x) \neq 0$

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Example Evaluate $\frac{dy}{dx}$ by product rule of differentiation. where

$$y = (4x^2-1)(7x^3+x)$$

Solu

$$\frac{dy}{dx} = (4x^2-1)\frac{d}{dx}(7x^3+x) + (7x^3+x)\frac{d}{dx}(4x^2-1)$$

$$= (4x^2-1)(21x^2+1) + (7x^3+x)\,8x$$

$$= 140x^4 - 9x^2 - 1 \quad \text{Ans.}$$

Example Evaluate $\frac{dy}{dx}$ of $y = \frac{x^3+2x^2-1}{x+5}$ by quotient rule of differentiation.

$$\frac{dy}{dx} = \frac{(x+5)\frac{d}{dx}(x^3+2x^2-1) - (x^3+2x^2-1)\frac{d}{dx}(x+5)}{(x+5)^2}$$

$$= \frac{(x+5)(3x^2+4x) - (x^3+2x^2-1)(1)}{(x+5)^2}$$

$$= \frac{2x^3+17x^2+20x+1}{(x+5)^2} \quad \text{Ans.}$$

where $x \neq 5$

* <u>practice sheet</u>: H. Anton.

chapter 1.2 → 31, 37, 38, 39, 40

" 1.5 → 11-22, 29, 30

" 2.2 → 9, 14, 46, 47-50

" 2.3 → 1-24

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